UNIVERSITÄT OSNABRÜCK

Programmieren für PsychologInnen

Verhaltensexperimente in Python - Teil I

Allgemeine Psychologie I
B.Sc. Psychologie / M.Sc. Psychologie - Seminar Programmieren
13.06.2025

Merle Sagehorn, M.Sc.

Agenda

25/04/25	Einführung in Python, Programmierlogik, Erste Basics
02/05/25	**Keine Präsenz-Session**
09/05/25	Variablen / Konditionen - If & Match
16/05/25	Schleifen - For & While / Textfiles
23/05/25	Statistik etc / Funktionen
30/05/25	**Keine Präsenz-Session:** Mini-Projektaufgaben
06/06/25	Plotting
13/06/25	Anwendungsbeispiel: Verhaltensexperiment I
20/06/25	**Keine Präsenz-Session:** Durchführung / Mini-Erhebung
27/06/25	Anwendungsbeispiel: Verhaltensexperiment II
04/07/25	Anwendungsbeispiel: EEG-Experiment I
11/07/25	Anwendungsbeispiel: EEG-Experiment II
18/07/25	Abschluss-Session

Aufgaben - Plotting

```
/var/folders/kj/bjcdpjl94s56jmvbtpqvdrkh0000gn/T/ipykernel_2425/3753346993.py:35: RuntimeWarning: invalid valu
e encountered in log
  axs[1,1].scatter(np.log(x), y, marker='.', color='limegreen')
```

- invalid value encountered in log → log(x) kann nur für positive Werte berechnet werden
- negative Werte und 0 lösen eine Fehlermeldung aus & können daher nicht geplottet werden

Today

- Vor der Durchführung
  - Experiment / Design
  - Präsentationsprogramm schreiben
- Durchführung
- Vor der Datenanalyse
  - Auslesen des Outputs
- Datenanalyse & Visualisierung

Verhaltensexperiment: Go/No-Go Task

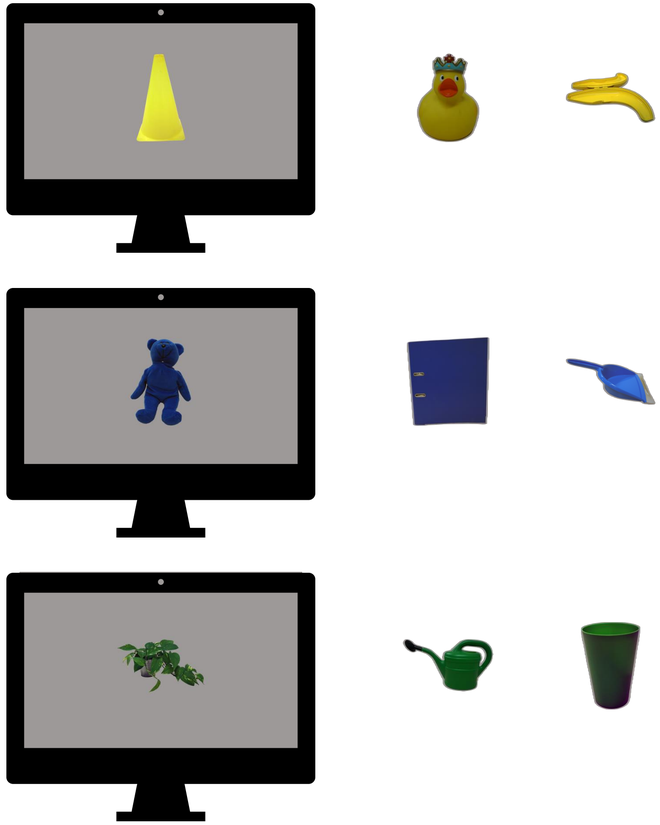
- = gängiges Verfahren zur Messung der Reaktionshemmung oder Inhibition

- Go-Bedingung: Reaktion
- No-Go-Bedingung: keine Reaktion
- —> Go Trials > No-Go Trials

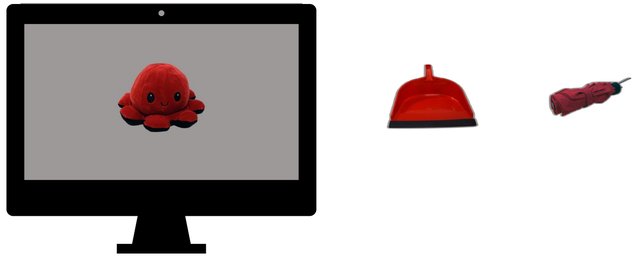
- Maß(e) zur Analyse:
  - Treffer- und Fehlerraten, Sensitivität vs. Spezifität, …

Verhaltensexperiment: Go/No-Go Task

GO-Trials

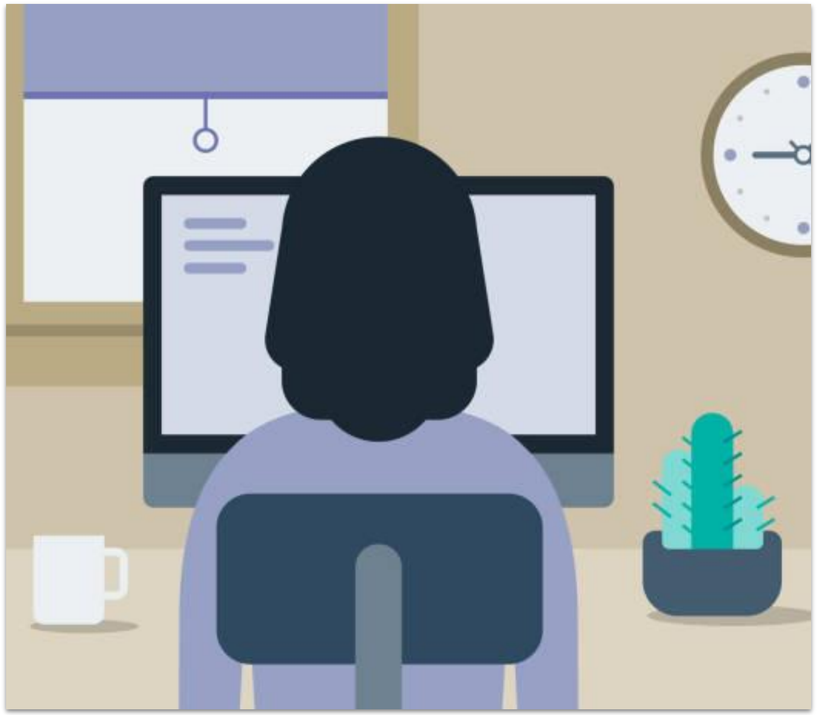


NOGO-Trials



Programm - Go/NoGo

- Was steckt hinter dem Programm?
  - Prinzip: Loop über die verschiedenen Stimuli—> Stimuli präsentieren —> Antworten erfassen & speichern
  - Wie werden die Stimuli präsentiert?
    - PsychoPy
  - Wie werden die Antworten gespeichert?
    - Erfassung des Tastendrucks & Messen der Reaktionszeit



PsychoPy

- Python Toolbox für PsychologInnen

- Hilfreiche Add-ons zum Erstellen von Experimenten in zwei Varianten:

  - als GUI-Builder mit grafischem Interface

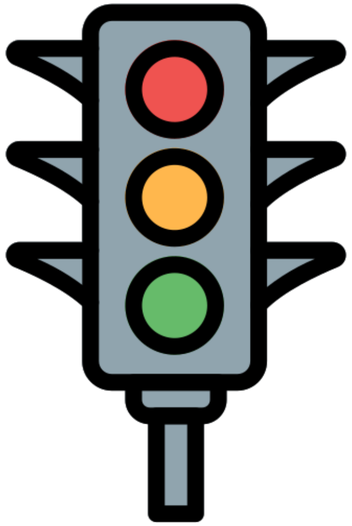
  - in Skriptform

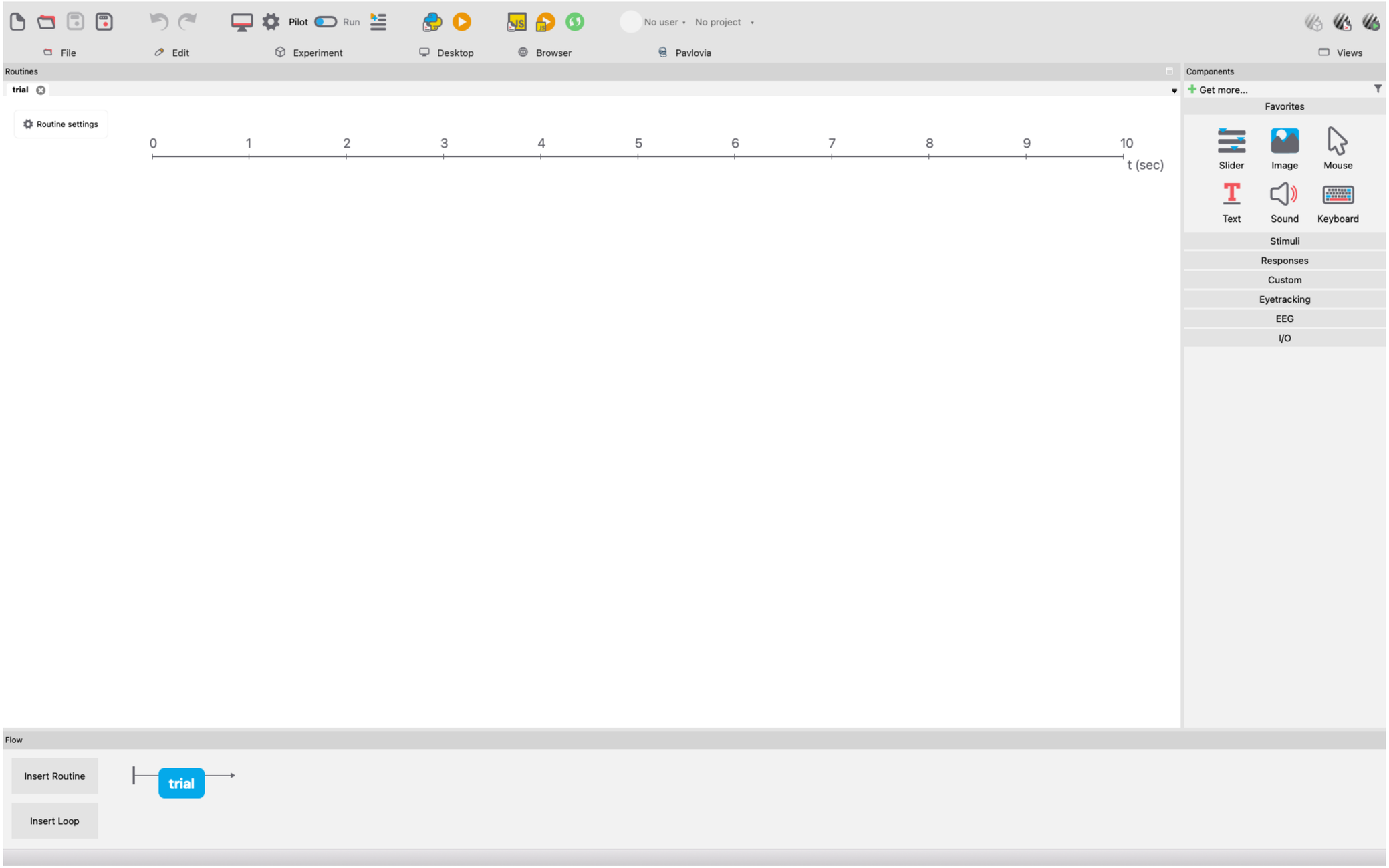
- vollständig open-source und mit Linux, Windows und mit Mac kompatibel

Website ↑

Download & Installation ↑

Let’s Go No-Go!

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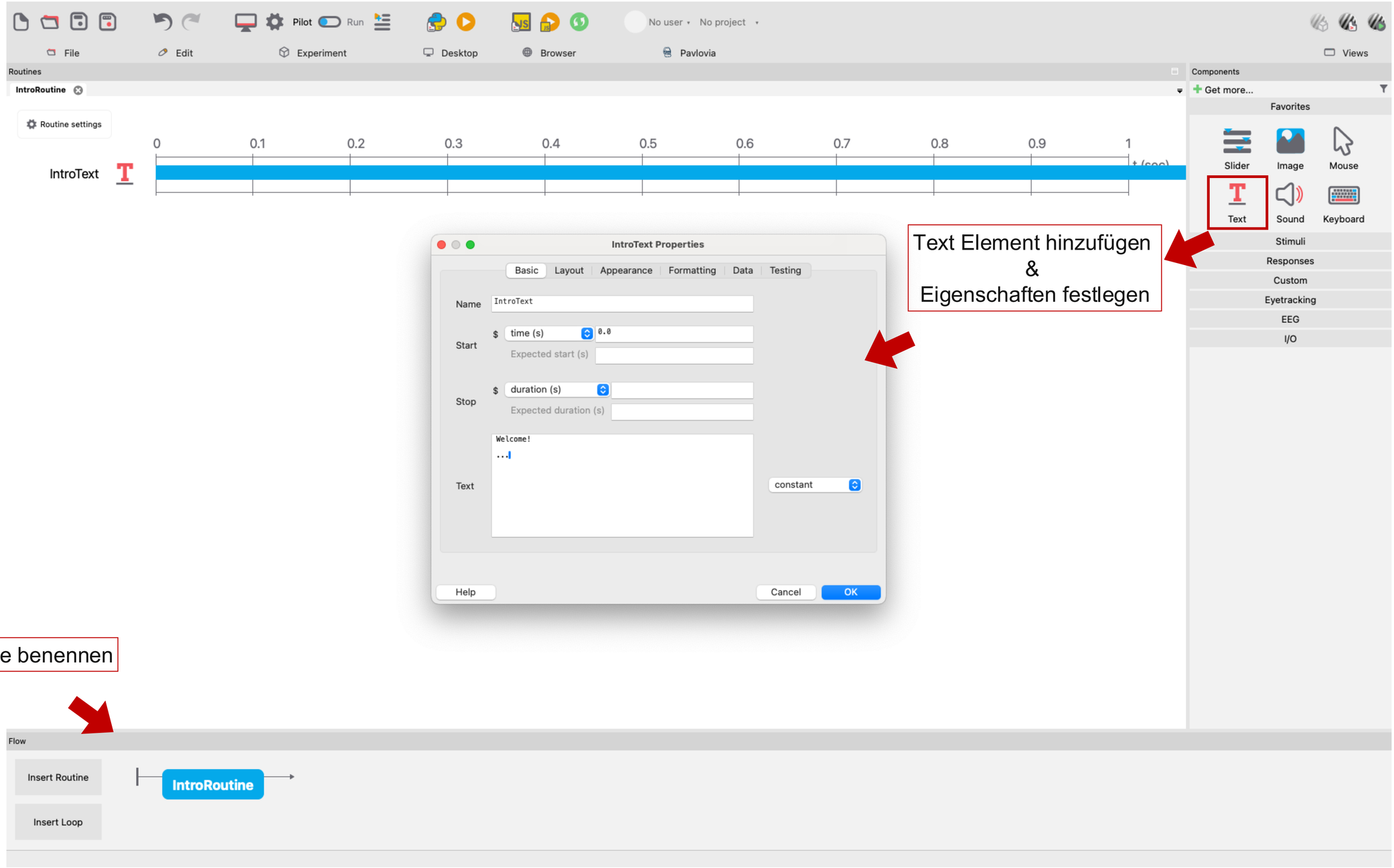
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Expected duration (s)
Text
Welcome!
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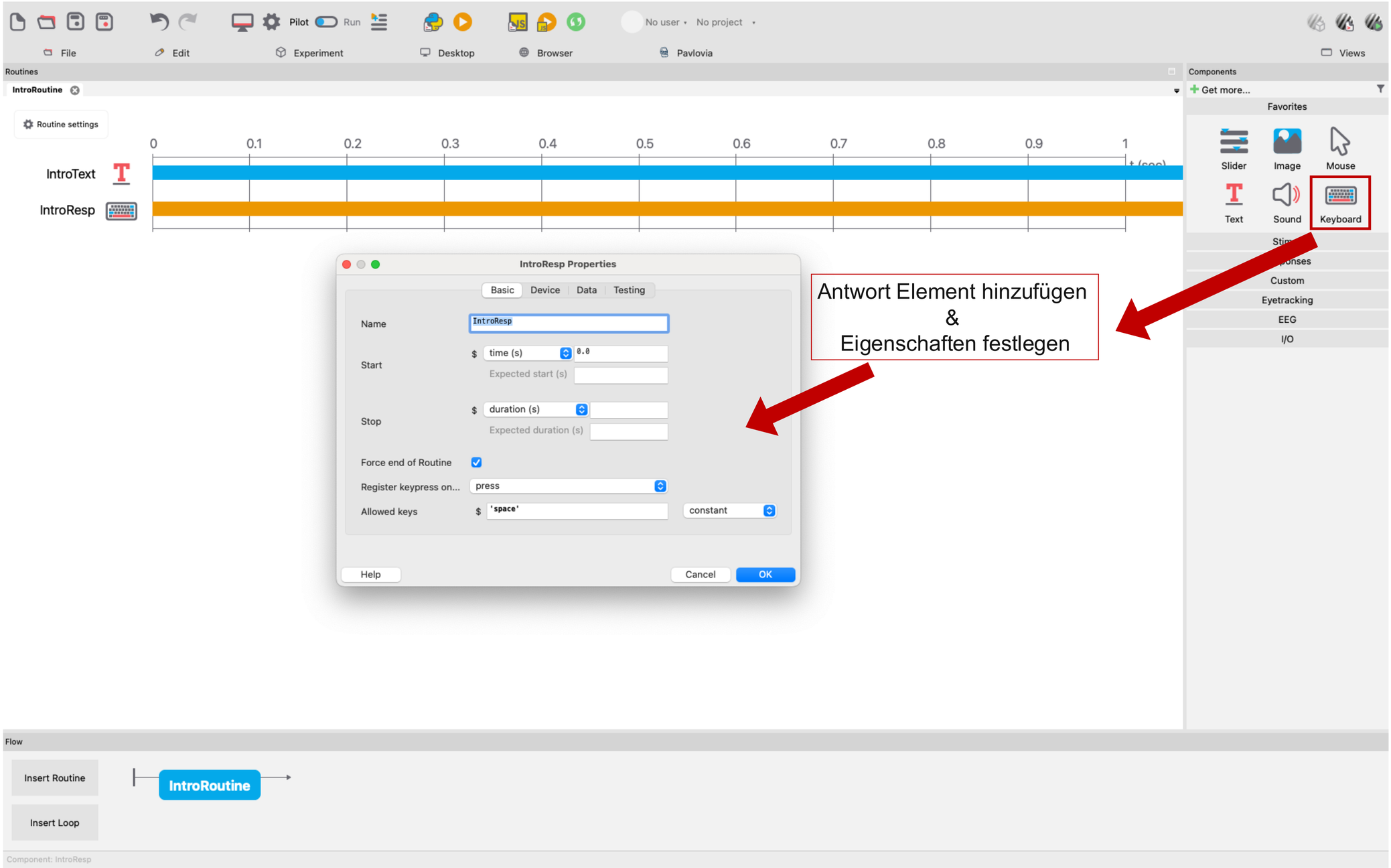
Help
Cancel
OK
Text Element hinzufügen
&
Eigenschaften festlegen
Routine benennen
Flow
Insert Routine
Insert Loop
IntroRoutine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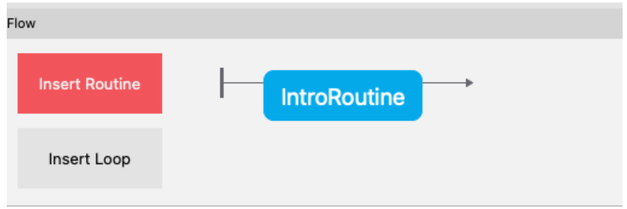
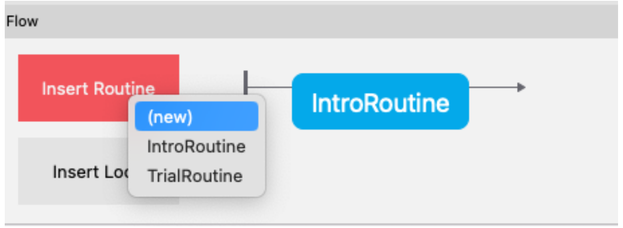
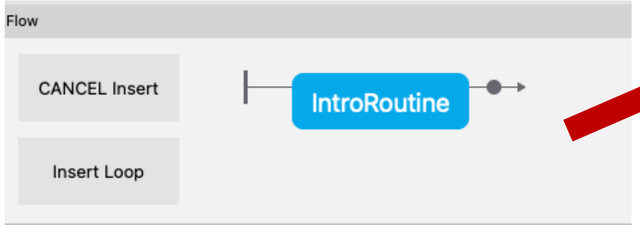
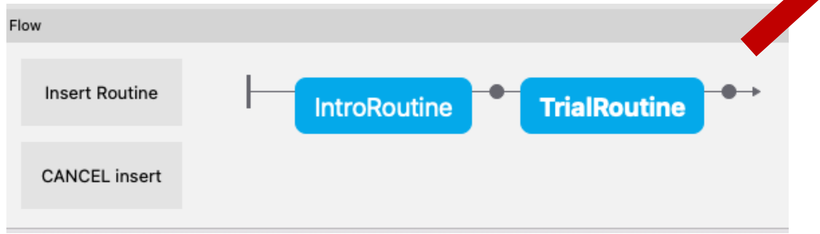
duration (s)
Expected duration (s)
Force end of Routine
Register keypress on...
press
Allowed keys
'space'
constant
Help
Cancel
OK
Antwort Element hinzufügen
&
Eigenschaften festlegen
Flow
Insert Routine
Insert Loop
IntroRoutine
Component: IntroResp

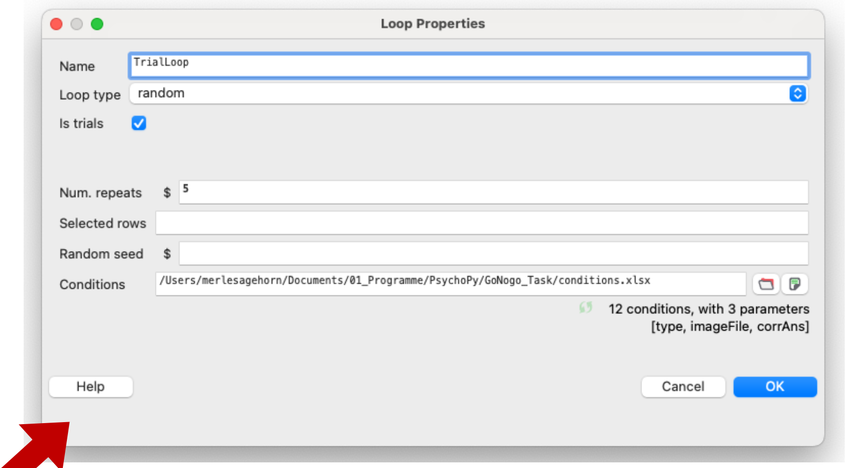
Trial Routine anlegen
Loop hinzufügen
Loop Eigenschaften definieren

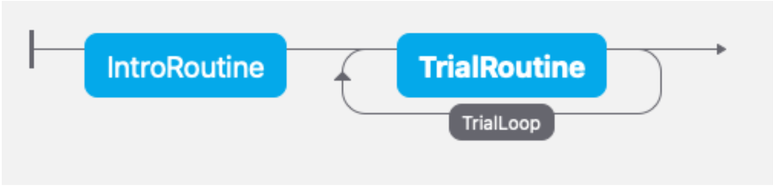


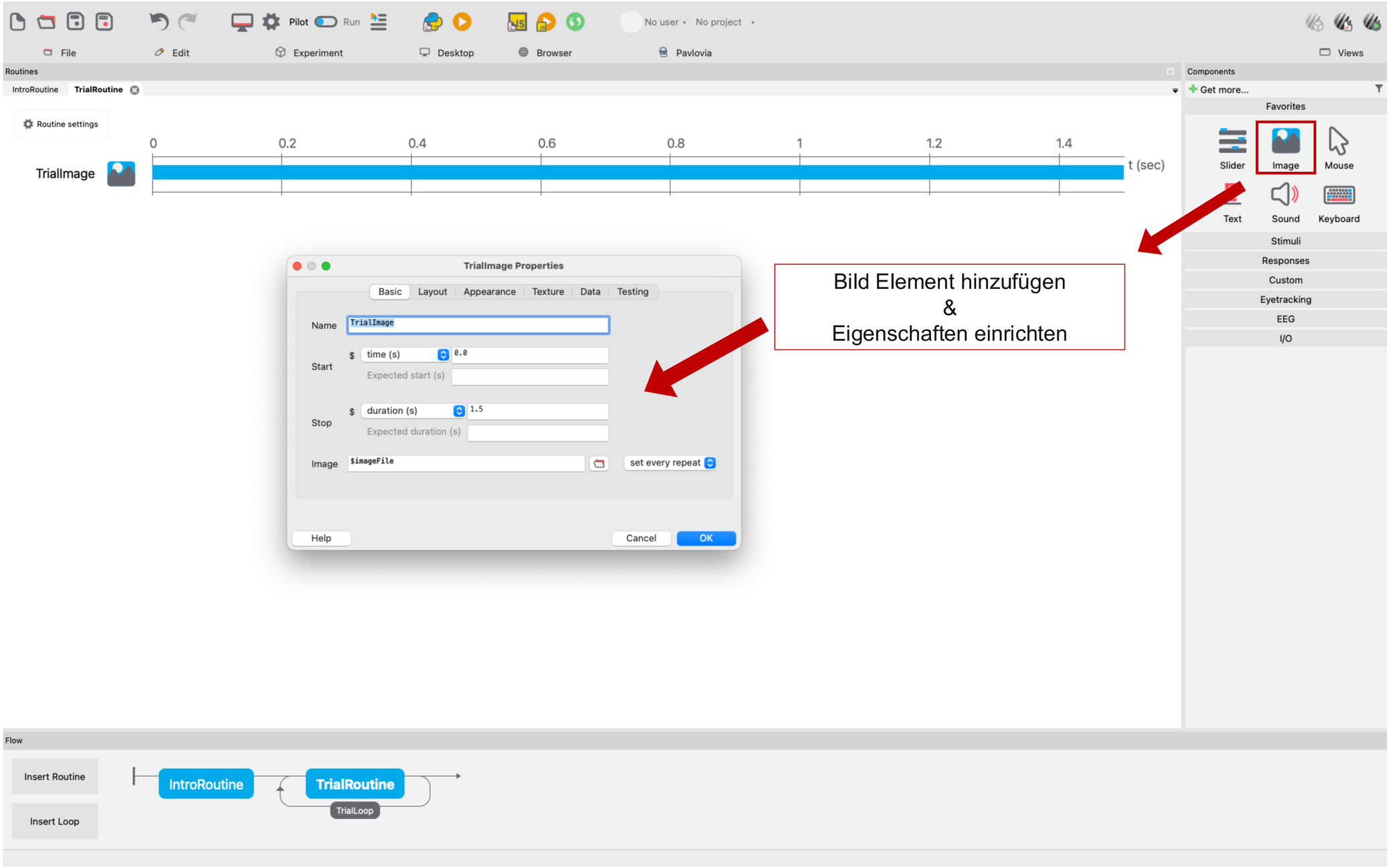








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set every repeat
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Bild Element hinzufügen
&
Eigenschaften einrichten
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Insert Loop
IntroRoutine
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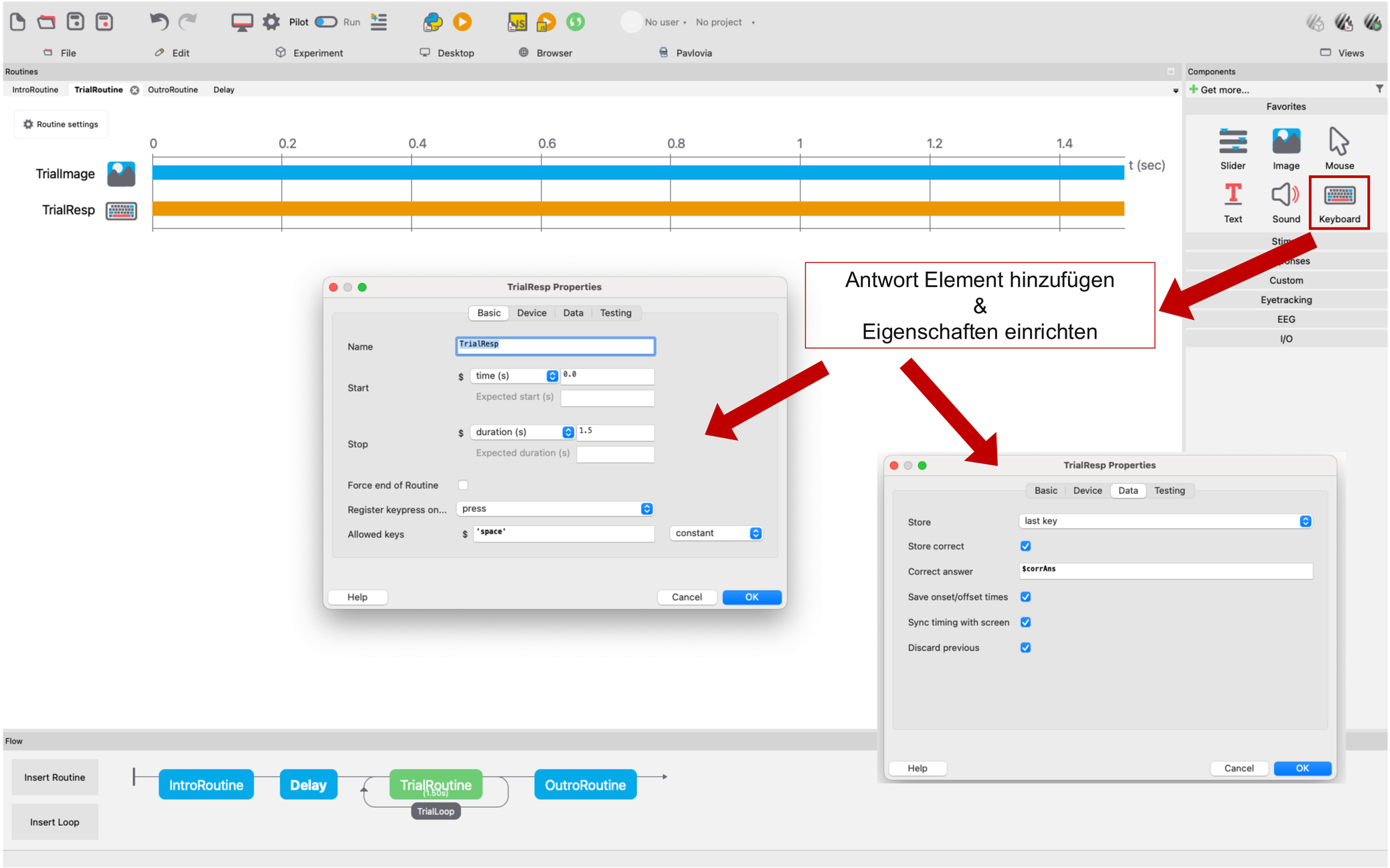
TrialLoop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duration (s)
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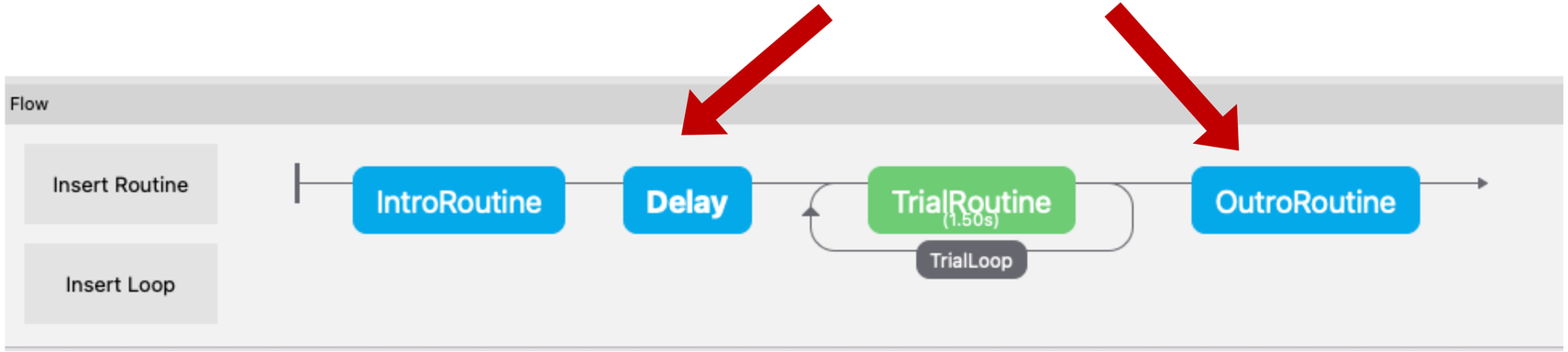
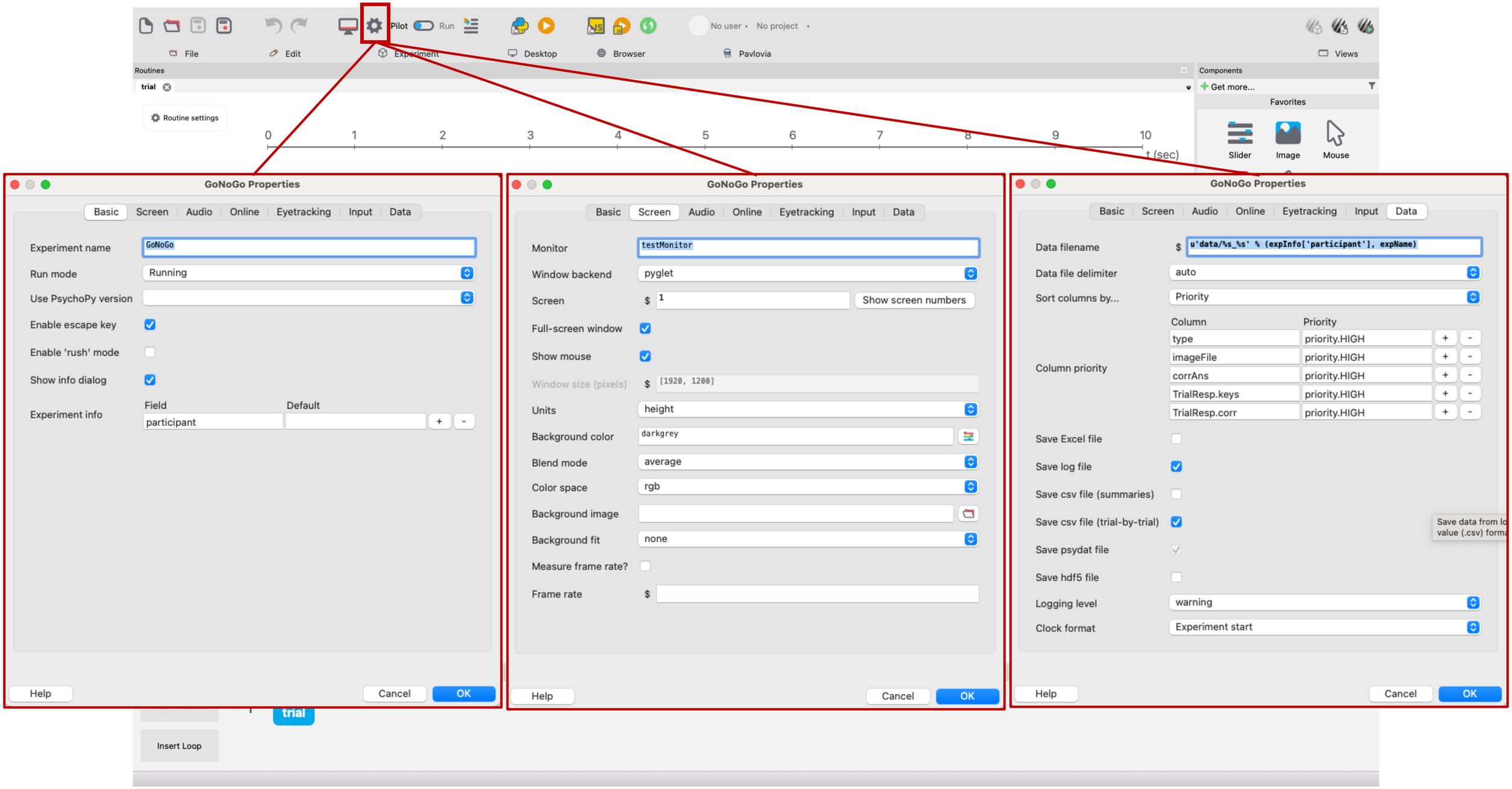
Expected duration (s)
Force end of Routine
Register keypress on...
press
Allowed keys
'space'
constant
Help
Cancel
OK
Store
last key
Store correct
Correct answer
$corrAns
Save onset/offset times
Sync timing with screen
Discard previous
Flow
Insert Routine
Insert Loop
IntroRoutine
Delay
TrialRoutine
(1.50s)
TrialLoop
OutroRoutine

Delay & Outro hinzufügen

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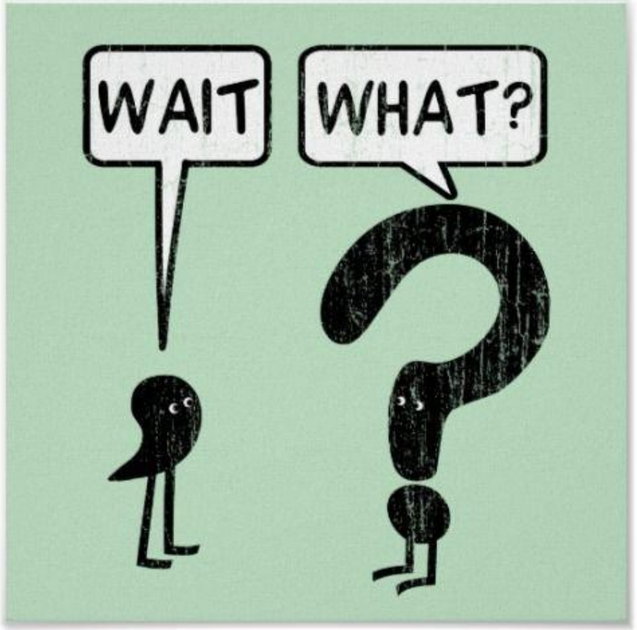
Use PsychoPy version
Enable escape key
Enable 'rush' mode
Show info dialog
Experiment info
Field
Default
participant
Help
Cancel
OK
Monitor
testMonitor
Window backend
pyglet
Screen
1
Show screen numbers
Full-screen window
Show mouse
Window size (pixels)
[1920, 1200]
Units
height
Background color
darkgrey
Blend mode
average
Color space
rgb
Background image
Background fit
none
Measure frame rate?
Frame rate
Data filename
u'data/%s_%s' % (expInfo['participant'], expName)
Data file delimiter
auto
Sort columns by...
Priority
Column priority
Column
Priority
type
priority.HIGH
imageFile
priority.HIGH
corrAns
priority.HIGH
TrialResp.keys
priority.HIGH
TrialResp.corr
priority.HIGH
Save Excel file
Save log file
Save csv file (summaries)
Save csv file (trial-by-trial)
Save psydat file
Save hdf5 file
Logging level
warning
Clock format
Experiment start
Insert Loop

Your turn!

- Aufgaben:

- Mit dem selbst erstellten Go-Nogo Programm ein paar Datensätze (mind. einmal selbst durchführen) erheben und hochladen, damit wir diese in der nächsten Session gemeinsam auswerten können.
  - Hochladen: Anwendung / 01_Verhalten / data



Referenzen

- Hemmerich, W. (2022). StatistikGuru: Go/No-Go-Aufgabe. Retrieved from https://statistikguru.de/lexikon/go-no-go-aufgabe.html
- https://www.psychopy.org/index.html